Deadline Curvature Enclosures and a Wider Quantum Latch Capture Window

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***Abstract*—At unchanged $\beta = 5/16$, finite-reference curvature enclosures and third-order remainders give conditional retained-channel error below $123/2^{23} < 2^{-16}$ throughout 16 ± 2^{-5} model time units, doubling the preceding halfwidth. Mixed bath commutators remain explicit. Uniform continuum values are compared separately; thermal continuum derivative transfer remains open. Exact acceptance arithmetic inherits the declared propagation, Gaussian and rounding assumptions. Physical promotion is zero.**

***Index Terms*—quantum control, open quantum systems, finite reservoirs, certified propagation, conditional error bounds**

I. Contract and evidence boundary

This addendum extends the source-bound MQM response-curvature packet [1], using the launch-transport and directed-parameter records [2], [3]. It does not reconstruct a device or a full-word compiler. The frozen parameters are $\Omega = 64$, $B = 16$, $\kappa = 8$, $\beta = 5/16$, $T = 16$, $h = g = 1$, and $\lambda^2 = 4\sqrt{2}$. The launch, error target, thermal covariance and error interpretation are unchanged. Engineering follows exposure to the parent results; a fresh candidate list $w = 2^{-k}$, $3 \le k \le 24$, was declared before the new thermal evaluation.

The finite Gauss bath has $M = 192$ mathematical modes. Its two-excitation sector has

$$D_2 = 3 + 4M + \binom{M+1}{2} = 19299. \tag{1}$$

These modes and coordinates are not physical qubits. The thermal response is a fixed-launch retained-state correction, not an arbitrary-input diamond-norm bound. All bounds below inherit the ideal continuum embedding, Gaussian remainder, directed parameter intervals and specified IEEE-754 operation model. External validation of those dependencies remains open.

II. Retained third commutator

Let $H = H_B + K$ on the finite source sectors, $C = [K, H_B]$, $D = [C, H_B]$, and $\mathrm{ad}_A X = [A, X]$. The finite response is $\Delta_{2,M}(t) = \mathrm{Tr}_B X(t)$, with $\|X(t)\|_1 \le 2\nu$. Here ν is the finite reference occupation trace, not a continuum photon-number bound. Bath-only outer commutators vanish under the partial trace. Jacobi gives the exact identity

$$\begin{aligned}\mathrm{Tr}_B \,\mathrm{ad}_H^3 X = \mathrm{Tr}_B \big(&\mathrm{ad}_K^3 X + 2\,\mathrm{ad}_C \,\mathrm{ad}_K X \\ &+ \mathrm{ad}_K \,\mathrm{ad}_C X + \mathrm{ad}_D X\big).\end{aligned} \tag{2}$$

For example, $\mathrm{ad}_{H_B} \mathrm{ad}_K X = \mathrm{ad}_K \mathrm{ad}_{H_B} X - \mathrm{ad}_C X$; moving the bath commutator outward under the trace generates the mixed terms in (2). No assumption of commuting interaction-picture operators is made.

The inherited local bound is $\|K\| < 14$. With $G^2 = 128/\pi$, the flat-band second and fourth moments are

$$\|\delta c\|^2 = G^2 B^2/3, \qquad \|\delta^2 c\|^2 = G^2 B^4/5. \tag{3}$$

Gauss nodes reproduce these polynomial moments. On $Q \le 2$, excitation exchange has norm at most $\sqrt{2}$ times its one-particle coefficient norm. Thus, using $\pi > 3$,

$$\|C\| < 256/3, \qquad \|D\| \le 4096/\sqrt{5\pi} < 4096/3. \tag{4}$$

Trace-norm contractivity for Hermitian inputs and $\|[A, X]\|_1 \le 2\|A\|\|X\|_1$ imply

$$\begin{aligned}\tfrac{1}{2}\|\Delta^{(3)}_{2,M}(t)\|_1 &\le \nu\left(8 \cdot 14^3 + 12 \cdot 14 \cdot \tfrac{256}{3} + 2 \cdot \tfrac{4096}{3}\right) \\ &= \tfrac{117056}{3}\nu.\end{aligned} \tag{5}$$

This all-time finite-reference bound does not transfer a derivative to the continuum.

III. Deadline thermal enclosure

All 192 covariance-weighted hot columns and the cold vector were freshly propagated by 128 steps of length $1/8$, degree 48. Separate real and imaginary CSR products have at most 196 terms per row; the source absolute matrix norm is below 64. The rounding model is round-to-nearest binary64, with parameter mismatch d_H, occupation mismatch d_n and step majorant

$$s = 32\,2^{-43}3^8 + \frac{8^{49}}{49!(1 - 8/50)} + 2^{-800}. \tag{6}$$

Set $e = 16d_H + 256s$. The directed parameter intervals include an added 2^{-40} Hamiltonian allowance. See the archived rational source for every numerator and denominator.

For evolved hot matrix W, read out the reduced weighted dyads directly:

$$\mathrm{Tr}_B\{2HWN(HW)^\dagger - H^2WNW^\dagger - WN(H^2W)^\dagger\}, \tag{7}$$

then subtract the finite occupation trace times the cold curvature. A separate polarization route is retained as a diagnostic. The exact rational half sum of absolute real and imaginary returned entries bounds the matrix trace norm, so acceptance does not use eigenvalue rounding.

The enclosure adds

$$\begin{aligned}E_C = \nu_U\big[&4 \cdot 64^2(2e + e^2) \\ &+ 4d_H(128 + d_H)(1 + e)^2\big] + 4 \cdot 64^2 d_n + 2^{-20}.\end{aligned} \tag{8}$$

The final slot bounds direct dyad arithmetic: a conservative 2^{28} primitive-operation budget gives $\gamma = 2^{-25}/(1-2^{-25})$. The exact receipt verifies

$$32\gamma\nu_U 4(64+d_H)^2(1+e)^2 + 16\,2^{-43}\nu_U(64+d_H)^2(1+e)^2 < 2^{-20}. \tag{9}$$

The budget exceeds the source dyad products and reductions. The absolute charge bounds their sums before cancellation; undocumented BLAS precision or execution behavior is not promoted as independently verified arithmetic.

The diagnostic raw half trace norm is about 2.8361×10^{-10}. Including (8), the certified conditional upper value is about $7.6057\times10^{-5} < 2^{-13}$. The larger certified value reflects conservative arithmetic and propagation allowances. With the inherited finite slope bound 2^{-20}, the thermal variation is

$$d_2(w) = 2^{-20}w + \tfrac{1}{2}2^{-13}w^2 + \tfrac{1}{6}\tfrac{117056}{3}\nu_U w^3. \tag{10}$$

IV. Cold launch-moment enclosure

The archived launch vectors obey $Hp_0 = p_1$, $H^2p_0 = p_0 + p_2$. If a_j denotes their live projection after transport, then

$$q''(16) = 2\|a_1\|^2 - 2\Re\langle a_0, a_2\rangle. \tag{11}$$

Parameter, propagation and centered embedding errors are charged afresh on $U = 129/8$. If N_j encloses the returned live norm, the unit-launch error is $e_0 = e + 84UR$, with

$$R = \frac{(4U)^{192}}{192!(1-4U/193)}, \qquad e_2 = 3e_0/2. \tag{12}$$

The native curvature correction is

$$2(2N_1e_0 + e_0^2) + 2(e_0N_2 + e_2N_0 + e_0e_2). \tag{13}$$

Exact rational roots enclose N_j. This gives $|q''(16)| < 2^{-14}$, while $|q'(16)| < 2^{-16}$ is rechecked. These are population derivative quantities, not the error threshold.

From $H^3p_0 = 2p_1 + \lambda p_3$, $\|H^3p_0\| < 13/4$. Put $A = 7/2048$, $s_1 = N_1 + e_0$, $s_2 = N_2 + e_2$, $h_3 = 13/4$, and

$$b_0 = A + s_1w + s_2w^2/2 + h_3w^3/6,$$
$$b_1 = s_1 + s_2w + h_3w^2/2, \quad b_2 = s_2 + h_3w.$$

Then $|q^{(3)}| \le 6b_1b_2 + 2b_0h_3 =: J_q(w)$ and

$$d_q(w) = 2^{-16}w + \tfrac{1}{2}2^{-14}w^2 + J_q(w)w^3/6. \tag{14}$$

The rational $7/2048$ bounds the square root of the vacuum ceiling; it has no identification with the separate $7/768$ capacity proposal.

V. Uniform value ledger and decision

Uniform continuum values are compared independently of finite thermal derivatives. Let $V = 64/31457265$, $r = 2^{-21}$, $y = 8U + 25/16$, and

$$\eta = \frac{4(128/3)r}{1-r}\frac{y^{384}}{384!(1-y/385)} + \frac{2(128/3)r^9}{1-r}. \tag{15}$$

The fresh vacuum response and covariance slots are $336U^3VR$ and $2U^2\eta$. With $x_{16} = 12433696189517/58512905602490430$,

$$x_w = x_{16} + 6V(16w + w^2/2), \qquad G_w = \frac{x_w^2}{2(1-x_w)} < 2^{-25}. \tag{16}$$

All declared candidates have $x_w < 1$. The full error majorant is

$$E(w) = 3/2^{18} + 2^{-128} + d_q(w) + 3/2^{26} + d_2(w) + 336U^3VR + 2U^2\eta + G_w. \tag{17}$$

Exact cross multiplication on the 22 frozen candidates selects $w = 2^{-5}$ as the largest passing dyadic halfwidth. At this value,

$$E(w) < 123/2^{23} < 2^{-16}. \tag{18}$$

Its diagnostic total is $1.4385964497733272\times10^{-5}$. Candidate 2^{-4} fails this majorant; it is not an all-time physical failure theorem. The display enclosure 123 was selected after evaluation; acceptance stayed fixed at 2^{-16}.

The halfwidth is twice the parent 2^{-6}, 2048 times the first 2^{-16} retention interval, and 2^{-9} relative to T. Equal numerical values of time and probability parameters do not identify their types. No pulse-off or temperature reduction was introduced.

VI. Audit, limitations and next gate

Six seeded $M = 4, 8$ third-commutator controls agree within 1.463×10^{-13}, two thermal curvature controls within 1.054×10^{-20}, and two cold controls within 2.013×10^{-15}. The $M = 192$ readout comparison differs by 2.060×10^{-18}. These are same-agent diagnostics, not independent review. A separate script audits 22 rational decisions.

The regenerable 55 MB checkpoint is excluded; its hash, source and frozen outputs are captured. Permanent retention, replenished bath, arbitrary-input diamond accuracy, native admission and hardware remain open. Next: enclose the deadline thermal third derivative with a fourth-order remainder, preserving separate continuum value comparisons and the open derivative-transfer gate.

References

[1] B. W. Mayes, "MQM response curvature," v0.1, immutable repository source ff55d95b820583149811d92135796215b9a004c5, development packet, 2026.

[2] B. W. Mayes, "MQM launch transport," v0.1, archived launch moments and propagation receipts, 2026.

[3] B. W. Mayes, "MQM response jet," v0.1, archived finite sector model and directed parameter intervals, 2026. See the release input hashes and review gates.